MILITARY SPECIFICATION

PSYCHROMETER, SLING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of sling psychrometer used for determining relative humidity of atmospheric air by means of wet and dry bulb temperature readings.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-585 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup, Paperboard.

MILITARY

MIL-L-10547 - Liners, Case, and Sheet, Overwrap, Water-vaporproof or Waterproof, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6685

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARD

ASTM C162 - Standard Definitions of Terms Relating to Glass and Glass Products.

(Copies may be obtained upon application to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

U.S. Department of Commerce/National Bureau of Standards

NBS Monograph 150 Liquid-In-Glass Thermometry

(Copies may be obtained upon application to National Bureau of Standards, U.S. Department of Commerce, Washington DC 20234).

3. REQUIREMENTS

3.1 Materials. Materials shall be of the highest quality throughout and shall be suitable in all respects for the purpose intended.

3.2 Design and construction. The sling psychrometer shall consist of two thermometers, mounting frame, handle, clips, wick and case. A psychrometer table is to be furnished with the case, twelve extra wicks and at least eight feet of cotton or nylon cord are to be furnished in the case. See Figure 1 for a typical design of the sling psychrometer.

3.2.1 Thermometers.

3.2.1.1 Design. The thermometers shall be cylindrical in shape and shall consist of a stem, expansion space, bulb and shall contain mercury as the indicator. The stem shall be made of standard glass thermometer tubing, as defined in NBS Monograph 150, and the bulb shall be made of "normal" clear glass, as defined in NBS Monograph 150. The stem shall be provided with a strip of opalescent glass, running the entire length of the back of the stem, subtending not less than 90 degrees of the circumference and a neck above each bulb of slightly smaller diameter than the bulb. Each thermometer shall be provided with a nib at the upper end to aid in preventing movement when mounted in the frame (see 3.2.2). One of the thermometers shall be provided with a wick (3.2.5) attached to the thermometer bulb at the neck. The thermometer length shall be 9 $\pm$ 1/8 inches.

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3.2.1.2 Range. The thermometers shall have a range of 20°F to 120°F with graduations of 1°F. Each incremental 10°F graduation, starting at 20°F, shall be numerically marked. Graduations and numerals shall be marked with permanent black pigment.

3.2.2 Frame. The frame shall be made of corrosion-resistant metal and shall be provided with two parallel longitudinal grooves for mounting the thermometers. The grooves shall not produce any strain on the mounted thermometers. The groove design shall position the thermometers so that in psychrometer operation air passing over the wet-bulb shall not pass over the dry bulb. Threaded holes or other provisions shall be provided for attaching the clips (see 3.2.4) used to secure the thermometers. The frame design shall include provisions for the attachement of the handle (see 3.2.3). The frame shall have a non-reflective finish.

3.2.3 Handle. The handle shall consist of a plastic or corrosion-resistant metal cylinder, at least 3-1/4 inches long and 1/2 inches minimum diameter, and a corrosion-resistant metal fitting for attachment to the frame. The fitting shall allow the frame to rotate about the handle. The fitting design shall allow the handle to recess against the back of the frame when not in use and shall provide a maximum angle of 90° between the handle and frame for psychrometer operation.

3.2.4 Clips. Two corrosion-resistant metal clips shall be provided to secure each thermometer to the frame. One clip shall secure the thermometer to the frame adjacent the upper nib and the second clip shall secure the thermometer to the frame 6 inches (minimum) below the upper clip. The clips shall secure the thermometers to the frame without distorting the thermometer.

3.2.5 Wick. The wick shall be of clean muslin or bleached cotton sewn or woven into a tube. The wick diameter shall enable the wick to fit snugly over the thermometer bulb and the length shall allow the wick to extend 3/8 inch beyond the bulb onto the stem and 1/2 inch below the bulb. The wick shall be securely tied above and below the wet-bulb thermometer bulb by means of cotton thread.

3.2.6 Case. A protective case with hinged or sliding cover shall be provided to protect the sling psychrometer when not in use. The case and cover shall be fabricated of corrosion resistant metal or plastic. The case with cover attached shall be 12-3/8 ± 1/4 inches long, 2 ± 1/4 inches wide by 1-1/8 ± inches high and shall accommodate the sling psychrometer, twelve extra wicks and at least 8 feet of cotton or nylon thread.

3.2.7 Psychrometric table. The psychrometric table shall be an easily-read table, providing at normal barometric pressure, the percent relative humidity corresponding to either the wet and dry bulb thermometer readings or the dry-bulb thermometer reading and the difference between the readings. The table shall cover 0 to 100 percent relative humidity range in the wet bulb temperature range of 20°F to 120°F. The psychrometric table dimensions shall be at least 4 inches by 9 inches and the table shall be made of plastic material. The table shall be marked with suitable waterproof ink and shall not be affected by washing with soap and water.

3.3 Thermometer accuracy test. The thermometers shall be accurate to ± 1.2°F when tested as specified (see 4.4.1).

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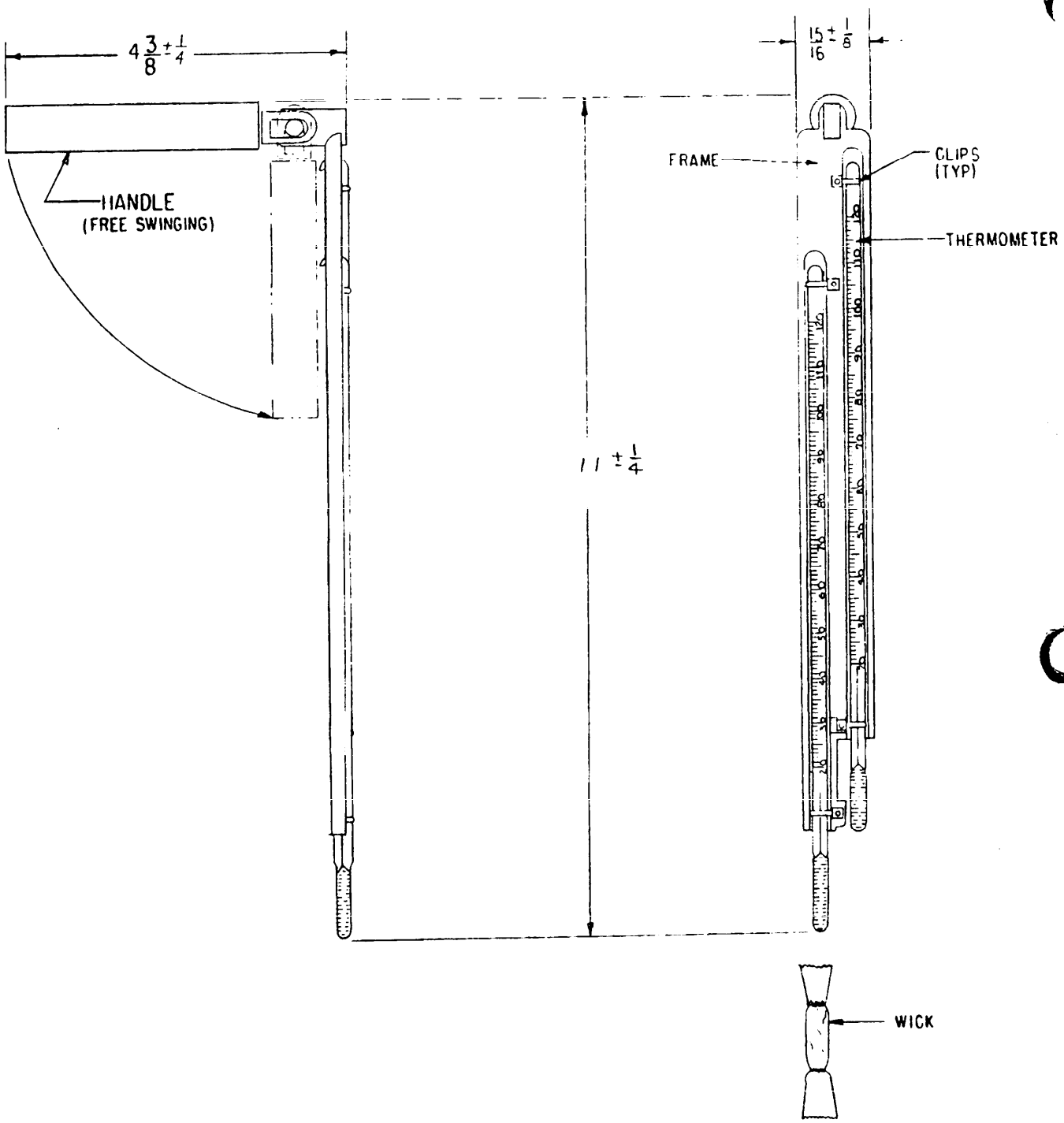


FIGURE 1. Psychrometer, Sling

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3.4 Identification marking. Each sling psychrometer frame and case shall be permanently and legibly marked with the name or trademark of the manufacturer.

3.5 Instructions. Instructions for use of the psychrometer shall be provided on the back of the psychrometric table. The instructions shall be printed with a suitable waterproof ink and shall not be affected when washing with soap and water.

3.6 Workmanship. Workmanship shall be first class throughout and the finished item shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the thermometer glass, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of product for sampling shall be one sling psychrometer.

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TABLE I. Sampling for examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. The inspection level shall be S-2 and the acceptance number shall be zero. The unit of product for sampling purposes shall be one sling psychrometer.

TABLE II. Sampling for tests.

Component	Characteristic	Requirement	Test Procedure
Sling Psychrometer	Thermometer Accuracy test	3.3	4.4.1

4.3 Examination. The sling psychrometer shall be examined to determine conformance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects listed in table III. Examination shall not be restricted to the classified possible defects listed below.

TABLE III. Classification of defects.

Categories	Defects
Major	
101	Component missing.
102	Component parts not free of burrs, sharp edges or corners.
103	Broken, cracked or distorted thermometer.
104	Thermometer mercury dull or dirty.
105	Case or cover cracked or deformed.
106	Thermometers not free of checks, chips, splits or protrusions.
107	Thermometer glass not clear.
108	Thermometer mercury column separated.
109	Thermometer scale not complete or legible.
110	Psychrometric table cracked or torn.
111	Psychrometric table information not complete, correct or legible.
112	Handle not free swinging.
Minor	
201	Marking incomplete, illegible or missing.
202	Components not free of dirt, grease or foreign material.

Examination shall not be restricted to the classified possible defects listed above. Glass defects are as defined in ASTM Designation C162.

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4.3.2 Dimensional examination. The sling psychrometer shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.3.3 Packaging inspection. The inspection of packaging, packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Thermometer accuracy. Thermometer accuracy shall be determined by comparing the thermometer readings with a thermometer certified to 0.1°F accuracy in the range of 20°F to 120°F in a liquid bath. Liquid bath temperature shall be held constant to 0.1°F. Accuracy shall be determined at 20°F, 32°F, 50°F, 80°F and 110°F.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Each complete psychrometer shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-676 or PPP-B-636, type CF, class domestic. Closure of box shall be as specified in the appendix of the applicable box specification.

5.1.2 Level C. Psychrometer shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Twenty-four (24) psychrometers shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproof as specified in the appendix of PPP-B-636.

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5.2.2 Level B. Twenty-four (24) psychrometers shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The packaged psychrometers shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking. Each unit package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The item covered by this specification is a sling psychrometer whose intended use is to determine relative humidity of atmospheric air.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number of item.
(c) Selection of applicable levels of packing and packing.

6.3 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the type normally procured to meet military requirements.

6.4 This specification covers the following item appearing in the Federal Supply Catalog.

National Stock Number	Item Identification
6685-00-438-2400	PSYCHROMETER , SLING

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DLA-DM

Project No. 6685-0594

Review activity:
DoD-MB

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

PSYCHROMETER, SLING MIL-P-37748

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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